

ACTIVE ZONE · DEL / VIDP · OPEN TRACK

ATC OMNI-ZONE

Terminal Safety System

Real-time collision prediction and runway-incursion detection for crowded terminal airspace.

TEAM CODE CRUSADERS

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THE PROBLEM

One controller, a sky full of simultaneous decisions

Controllers manually project where every aircraft will be next while also watching runway traffic — a workload that scales with every extra blip on the scope.



High cognitive load

Mentally tracking closure rates and separation for dozens of aircraft at once, in real time.



Mid-air collision risk

Manual projection leaves little margin when two aircraft converge quickly in busy terminal airspace.



Runway incursions

Ground traffic and stopping distances are just as easy to misjudge, especially with weather in play.



From the mission brief

“Air traffic controllers must manually predict future positions of aircraft in crowded terminal airspace while also monitoring runway traffic — creating a high cognitive workload and increasing the risk of mid-air collisions and runway incursions.”

THE SOLUTION

A live safety layer for the tactical map

ATC Omni-Zone ingests live ADS-B telemetry and weather, runs it through a physics-based conflict engine, and streams resolution advisories straight onto a tactical map — before a human has to spot the problem.



Live ADS-B + weather ingestion, with an offline mock fallback



Physics-based conflict engine for air and ground



React-Leaflet tactical map with geofence & runway geometry



Resolution advisories with live countdown timers



Concept view — active scenario management

60s

time-to-closest-
approach window

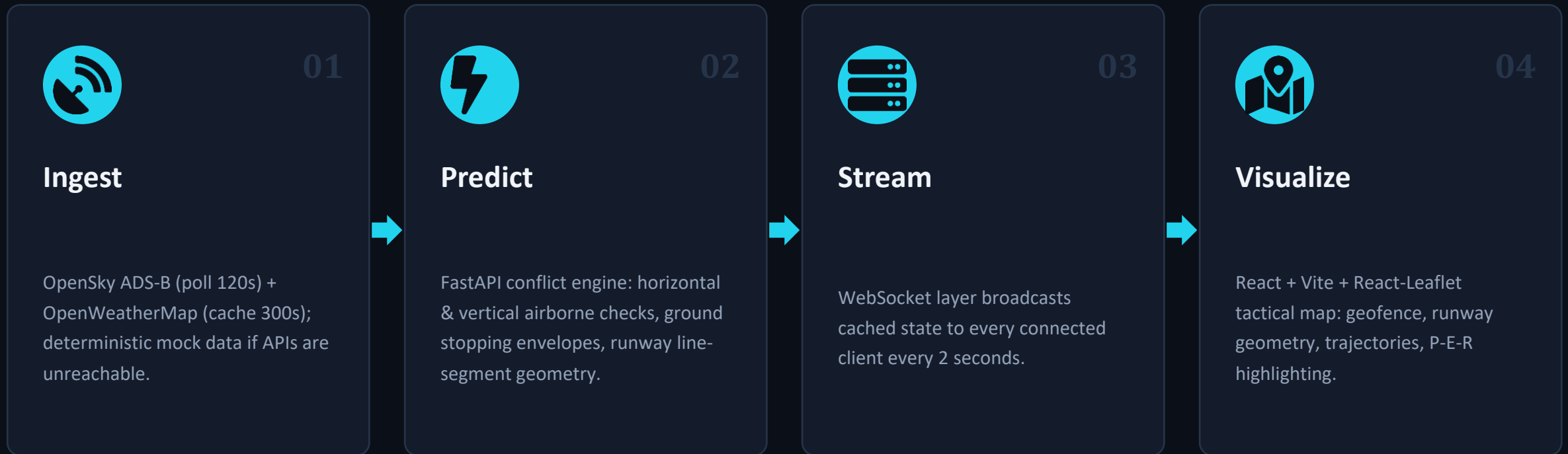
2s

cached telemetry
broadcast interval

50km

geofenced tactical
map radius

From raw telemetry to a resolution advisory



Known limitations: single fixed WebSocket route (DEL / VIDP), JSON airport config with in-memory caching instead of PostgreSQL/PostGIS, and a linear prediction model that doesn't yet model pilot response, ATC constraints, or terrain.

The tactical display, live



Resolution advisories

Plain-language RA text per aircraft — climb, descend, or turn — with a live countdown to closest approach.



P-E-R highlighting

Preemptive-Event-Red highlighting flags the exact aircraft driving each conflict on the map.



Aircraft inspection

Search by callsign or ICAO and drill into altitude, speed, heading, and a short trajectory forecast.

TECH STACK

Built on IBM technology, wrapped in a live map



IBM Bob

The IBM technology track requirement for this build, used alongside our own physics-based prediction engine to power the safety-analysis pipeline.

WHY IT FITS

Keeps the safety-critical math and data plumbing on a dependable, well-supported foundation so the team could focus hackathon time on the conflict-detection logic itself.

LANGUAGES

Python · JavaScript

BACKEND

FastAPI

FRONTEND

React · Vite

MAPPING

React-Leaflet

PROUDEST OF

The pure-mathematics collision engine: separate horizontal/vertical airborne checks, realistic forward stopping envelopes for ground traffic, runway line-segment geometry, and actionable resolution advisories instead of generic warnings.

What the dashboard catches—in one live session

22

aircraft tracked simultaneously

13

airborne, tracked in real time

3

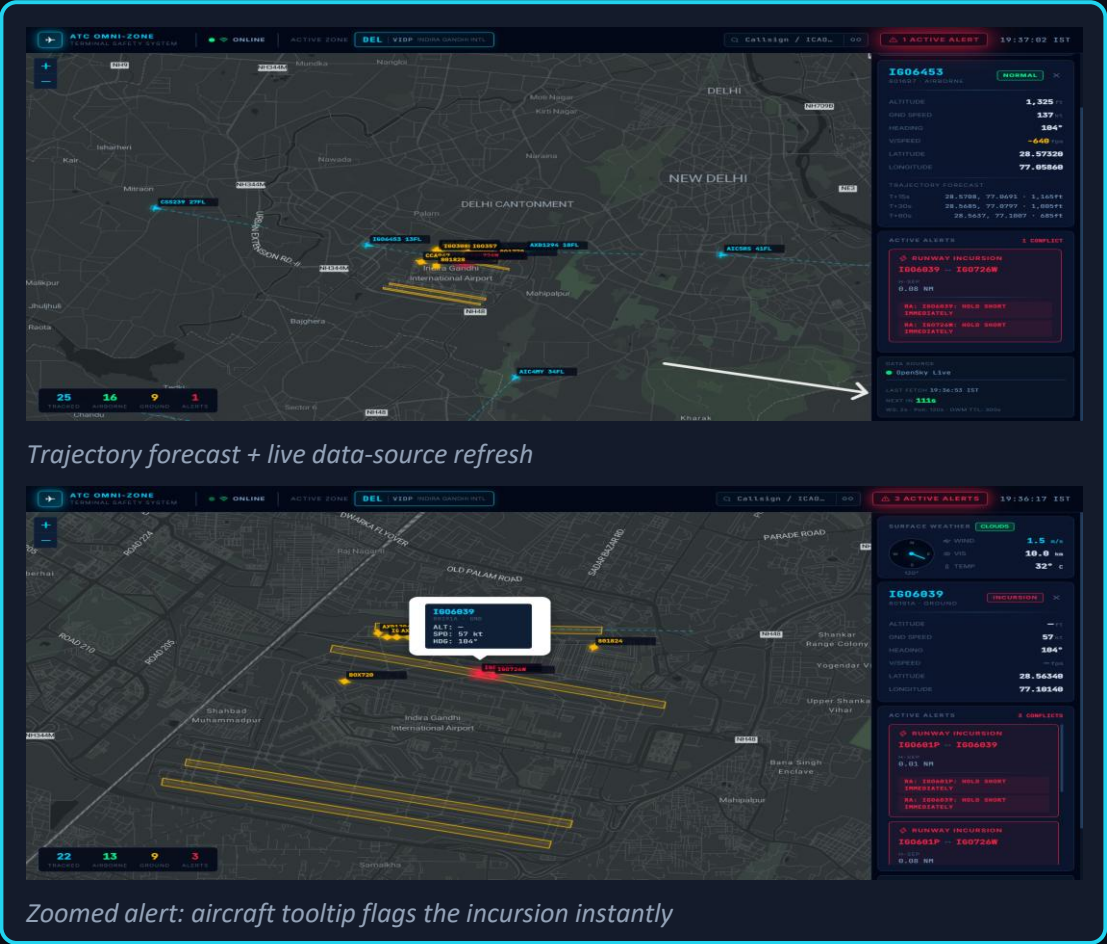
active conflicts flagged live

100%

UI uptime even if live APIs are down

Captured from a live run over Indira Gandhi International (DEL / VIDP): a runway incursion between IG0601P and IG06039 is caught and paired with an instant “HOLD SHORT” advisory for both aircraft — while the trajectory forecast keeps every other aircraft in view.

- Turns raw feeds into an actionable, countdown-driven alert — not just a warning light
- Same safety logic covers airborne separation and ground / runway conflicts
- Offline mock fallback keeps the tool demo-ready without live API access



THE TEAM

Code Crusaders

Open Track · ATC Omni-Zone Terminal Safety System



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